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Machine Learning and Deep Learning (Deep Learning)

An Explainable Deep Learning Framework for Breast
Cancer Classification from Histopathological Images
Using CNNs and Vision Transformers.

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1 MOTIVATION AND RATIONALE

Breast cancer is one of the most prevalent and deadly illnesses of women in the world. Early and timely diagnosis is a key factor in increasing survival rates and treating less complicated. This intractable manual diagnosis is, however, time-consuming and laborious and relies very much on the expertise of pathologists, with a high risk of misdiagnosis. The emergence of digital pathology has allowed the development of large facial-histopathological image datasets like BreakHis and make a development of machine learning and deep learning techniques possible to automatically classify the dataset in cancerous and non-cancerous images. These techniques have been found to be highly effective in aiding doctors with their diagnosis by enhancing accuracy and productivity. Though tremendous advancement has occurred in the field of breast cancer image classification, the field is still confronted with a number of problems. This task is very complicated due to the differences in tissue structure, staining and the high degree of intra-class similarity of both benign and malignant samples. Moreover, most of the already developed models are, unfortunately, not interpretable, and in medical applications, transparency in decision-making is needed. Therefore, to address these challenges motivating this study, the framework of comparative deep learning for the classification of breast cancer by histopathological image is proposed. It also compares several cutting-edge architectures such as CNN-based models and Vision Transformers to find the most effective model to accurately classify. Moreover, explainable AI methods like Grad-CAM++ are added to explain the interest areas in the histopathology pictures, adding to the intuitiveness and legitimacy of clinical choices.

The goal of this work is to help create a reliable and efficient computer aided diagnostic system that will assist Pathologists in identifying early stages of breast cancer and therefore be able to increase patient survival.

2 STATE OF THE ART (SOTA)

Over the past few years, there has been tremendous progress in breast cancer diagnosis using histopathological images, largely instigated by the advent of deep learning techniques. The hand-crafted features were used for the breast cancer classification using the traditional machine learning classifiers like Support Vector Machines (SVM), K-Nearest Neighbors (KNN) and Random Forest (RF) classifiers. But these approaches only capture the complex spatial pattern from histopathology images to a limited extent [1]. Convolutional Neural Networks (CNNs) have emerged as the most common solution to medical image analysis with the advent of deep learning. Superior results have been achieved in the automated breast cancer classification tasks by architectures like AlexNet, VGGNet, ResNet, and DenseNet. Of these, DenseNet demonstrates this ability to reuse features with the best result, which is especially useful for medical imaging applications [2]. Recently iEfficientNet and Xception are introduced which were efficient architectures and were improved because of the improved accuracy with less computational cost. The models are based on optimized Convolutional structures to improve fine-grained tissue patterns in the histopathological images. Meanwhile, a novel paradigm known as Vision Transformers (ViT) has been introduced in medical image classification. At the same time, another approach called Vision Transformers (ViT) has been proposed as an alternative paradigm in the field of medical image classification. Unlike CNNs, ViT models leverage their self-attention properties for capturing global dependencies, allows models to compete effectively with the CNN in large-scale datasets [3]. These developments notwithstanding, many models currently available are often scarcely interpretable, which is essential in clinical applications. To tackle this, concepts of explainable AI (xAI) like Grad-CAM and Grad-CAM++ have been developed to show location of attention of the model, enhancing transparency and trustworthiness of prediction system.

Nevertheless, some obstacles in using optimal performance on datasets such as BreakHis are still encountered due to three circumstances: a) class imbalance, b) inter-patient variability, and c) the subtle difference between the benign and malignant tissues. This induces strong demand for

the study of several deep learning and transformer-based models that can be compared a robust classifiability of breast cancer.

3 OBJECTIVES

The main goal of this research is to build an interpretable and a valid deep learning system for classification of breast cancer, based on histopathological images. The particular aims of this research are given below:

- **Designing an Automated Classification System** – To design a deep learning framework which is able to differentiate between benign and malignant breast cancer histopathology images.
- **Compare State-of-the-Art Models** – to compare many of these architectures including CNN-based models and Vision Transformers like DenseNet121, EfficientNetB0, ResNet50, Xception, MobileNetV3 and ViT-B/16.
- **Enhance With Transfer Learning and Fine-tune On BreakHis** – To better classify the BreakHis dataset by fine tuning and overall transfer learning.
- **Enable Model Interpretability** – To enable methods known as explainable AI (Grad CAM++) to visualize the regions of interest (ROI) in histopathological images.
- **Minimise False Negatives** – To maximise recall in model evaluation, meaning to not misclassify any negative cases (malignancy), failures should be minimised.
- **Support Clinical Decision Making** – To help healthcare workers capture accurate cancer diagnostics predictions, which are interpretable by them.
- **Set up a Benchmark Comparison** – to have a full comparison of the performance of deep learning and transformer-based models as a future benchmarking.

4 METHODOLOGY

Comparative Analysis of Deep Learning and Vision Transformer Models for Breast Cancer Classification Using Histopathological Images

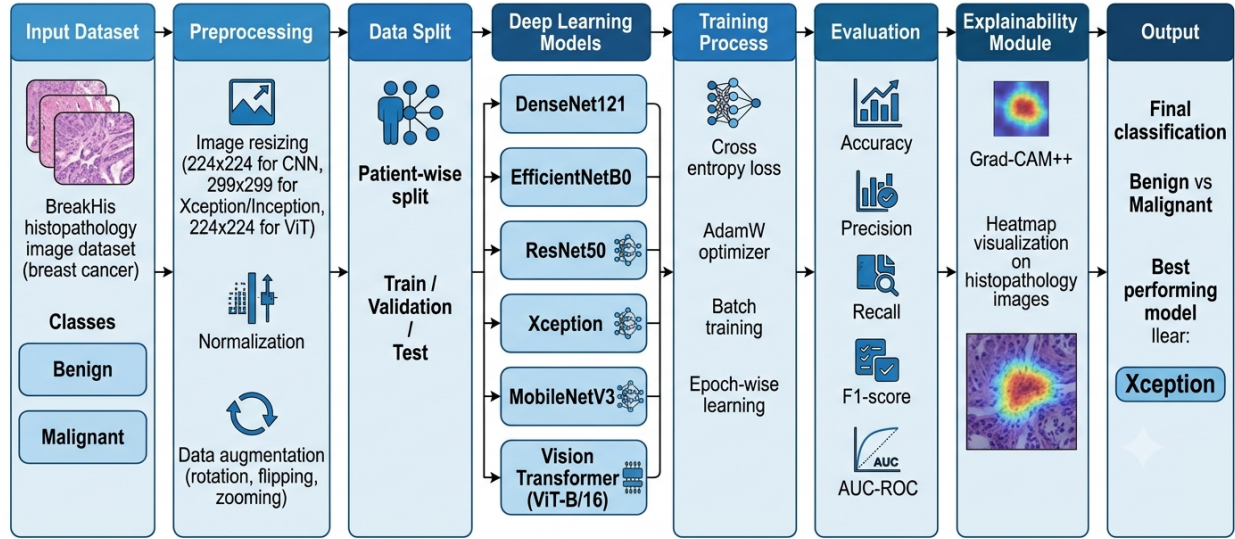


Figure 4.1: Proposed methodology for breast cancer classification using deep learning models and explainable AI.

To accomplish these goals, a structured deep learning pipeline has been created for breast cancer classification of histopathological images from the BreakHis dataset. Overall, the workflow can be divided into four major steps: Dataset preparation, preprocessing, model training, evaluation and explainability analysis.

4.1 Dataset Collection

This study is based on the BreakHis dataset which includes histopathological images from breast tumour tissues. The data set is divided into either of the classes, “benign” or “malignant.” They take photographs of the image at different magnification levels, and train the model with those

images they captured and test it on the ones they didn't. At different magnifications, the images are taken at different Zoom levels and they use the images to train and test their model at different magnifications [4].

4.2 Data Preprocessing and Splitting

All images are reduced to a fixed size suitable for Deep Learning models – 224×224 for CNN based models, 299×299 for Xception/Inception models. The Pixel values are scaled by their ImageNet statistics to obtain a stable training. We use data augmentation methods like rotation, flipping and zooming to enhance generalization of the data.

A balanced dataset of training, validation and test sets is created, which is done to prevent data leakage and provides for a proper evaluation.

4.3 Model Architectures

With this study, various types of deep learning architectures are tested to see and contrast their performance on breast cancer classification [5]. These include:

- **DenseNet121:** DenseNet121 is a dense CNN with dense connections from one layer to another, which allows for the reuse of features and can facilitate the smooth flow of information from one layer to another. It is an architecture that minimizes the parameters, enhances feature propagation and classification performance.
- **EfficientNetB0:** EfficientNetB0 is a lightweight and efficient CNN, which scales up network depth, width and resolution in a balanced manner. It performs well with less number of parameters than conventional CNN architectures.
- **ResNet50:** ResNet50 is a Deep Residual Neural Network that has 50 layers. It helps to resolve the vanishing gradient problem with residual connections: it is a way of making deeper and better networks.
- **Xception:** Xception is an advanced Convolutional Neural Network (CNN) that is designed using a technique called depthwise separable convolutions. It enhances the abilities of fea-

ture extraction and has shown excellent performance on several medical image classification problems.

- **MobileNetV3:** MobileNetV3 is a recently published lightweight neural network for low memory devices. It incorporates squeeze and excitation modules and depthwise separated convolutions along with the neural architecture search to deliver high accuracy at low computation.
- **Vision Transformer (ViT-B/16):** Vision Transformer (ViT-B/16) extends the use of the transformer model to the domain of image classification, segmenting images into fixed-size patches, with the help of self-attention mechanisms. It provides a good grasp of the image relationship at the local and global level.

Pre-trained images of ImageNet are used as the initial conditions for each model and fine-tuned on the BreakHis database.

4.4 Training Strategy

Cross-Entropy Loss function is used for all models and AdamW optimizer for optimization. Weight decay is used for regularization and the learning rate is $1e-4$ [6], [7]. Optimizing is optimised by training several epochs, expanding on individual training batches.

To keep the best model according to the validation result, early stopping and model checkpointing are adopted.

4.5 Evaluation Metrics

Several measures of model performance are utilized such as accuracy, precision, recall and F1-score, as well as the Area Under the ROC Curve (AUC). These metrics perform a comprehensive assessment of classification performance, particularly in medical diagnosis applications, in which recall is key.

4.6 Explainability Using Grad-CAM++

Based on the aim of interpretability of models, Grad-CAM++ is used to visualize the region of the histopathological images that contribute most towards the model's predictions. This becomes useful in comprehending decisions made by the model and to ensure trust in the clinical use [8].

Heat-maps generated highlight important tissue structure and regions of abnormality that can affect the classification outcome.

5 EXPERIMENTS AND RESULTS

The setup of experiments, performance comparison and measurement of various deep learning and transformer models for the classification of breast cancer images from the BreakHis histopathological image database are described. The experimental setup and performance comparison/measurement of multiple deep learning and transformer-based models in breast cancer classification using the BreakHis histopathologic image database are provided. Here are the models that have been evaluated: DenseNet121, EfficientNetB0, ResNet50, Xception, MobileNetV3, and Vision Transformer (ViT-B/16).

5.1 Experimental Setup

Experiments were all carried out on a Python and GPUs programming interface called PyTorch. The weights for all Models were frozen using the pre-trained ImageNet weights and fine-tuned the model on the BreakHis dataset.

The models were trained using the following configuration:

- Loss Function: Cross-Entropy Loss
- Optimizer: AdamW
- Learning Rate: 1×10^{-4}
- Batch Size: 16 or 32 (based on GPU memory)
- Epochs: 10
- Image Size: 224×224 (CNNs), 299×299 (Xception)

5.2 Evaluation Metrics

All models were tested with regard to Accuracy, Precision, Recall, F1-score, and Area Under the curve (AUC). These metrics enable a full measurement particularly for medical classification applications which have a high dependence on the absence of a false negativ

5.3 Comparative Results of Models

The results of the experiment are presented in column 2 to compare the different architectures on the BreakHis dataset. The performance summary is showed as follows:

Table 5.1: Performance Comparison of Deep Learning Models

Model	Accuracy	Precision	Recall	F1-score	AUC
Xception	0.8148	0.8199	0.9140	0.8644	0.8879
DenseNet121	0.7946	0.7922	0.9244	0.8532	0.8723
MobileNetV3	0.8005	0.8162	0.8918	0.8523	0.8625
ResNet50	0.8005	0.8162	0.8918	0.8523	0.8625
EfficientNetB0	0.8005	0.8162	0.8918	0.8523	0.8625
ViT-B/16	0.7500	0.7560	0.9048	0.8237	0.8212

5.4 Analysis of Results

5.4.1 DenseNet121

DenseNet121 achieved an accuracy of 79.46%, precision of 79.22%, recall of 92.44%, F1-score of 85.32%, and AUC of 87.23%. The model exhibited high learning stability and good recall rate for malignant cases, enabling good feature reuse by important dense connectivity.

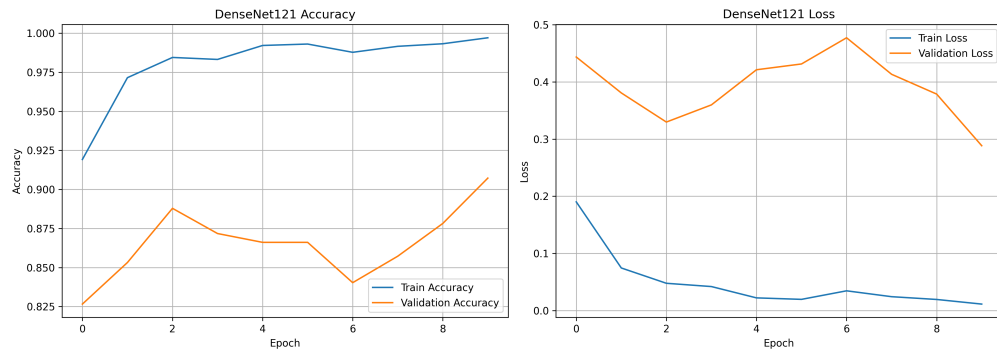


Figure 5.1: Training and Validation Accuracy/Loss Curves of DenseNet121

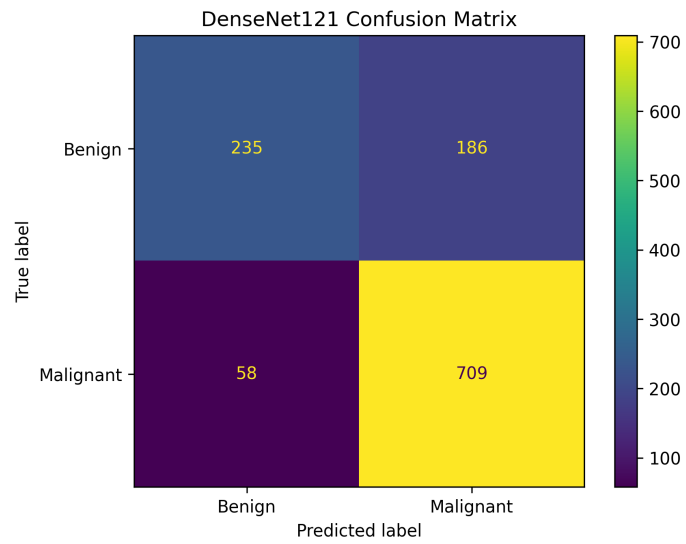


Figure 5.2: Confusion Matrix of DenseNet121

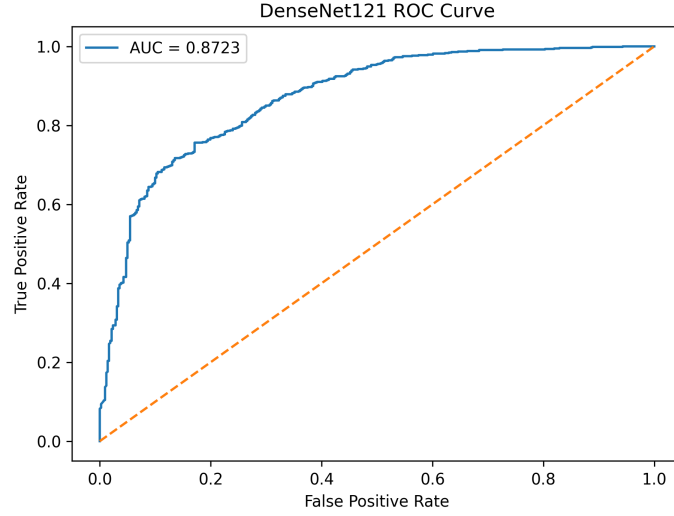


Figure 5.3: ROC Curve of DenseNet121

5.4.2 EfficientNetB0

EfficientNetB0 achieved an accuracy of 80.05%, precision of 81.62%, recall of 89.18%, F1-score of 85.23%, and AUC of 86.25%. The model was well-balanced in terms of performance and computational requirements.

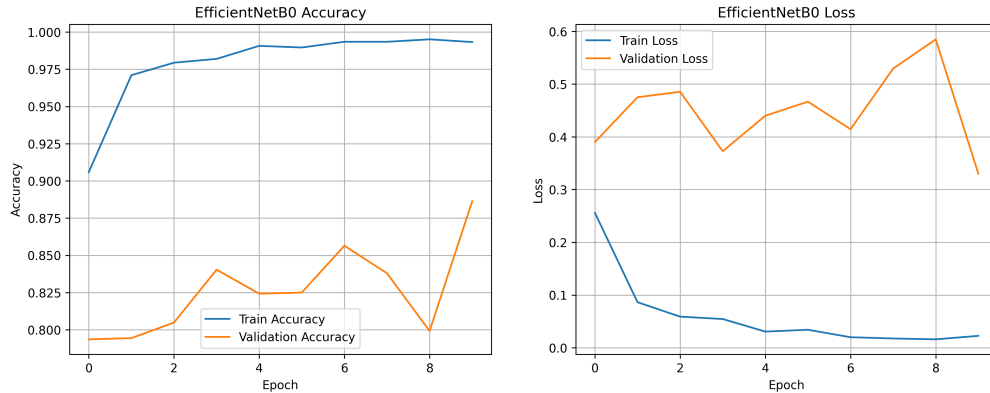


Figure 5.4: Training and Validation Accuracy/Loss Curves of EfficientNetB0

5.4.3 ResNet50

ResNet50 achieved an accuracy of 80.05%, precision of 81.62%, recall of 89.18%, F1-score of 85.23%, and AUC of 86.25%. Residual connections did help in alleviating the vanishing gradient

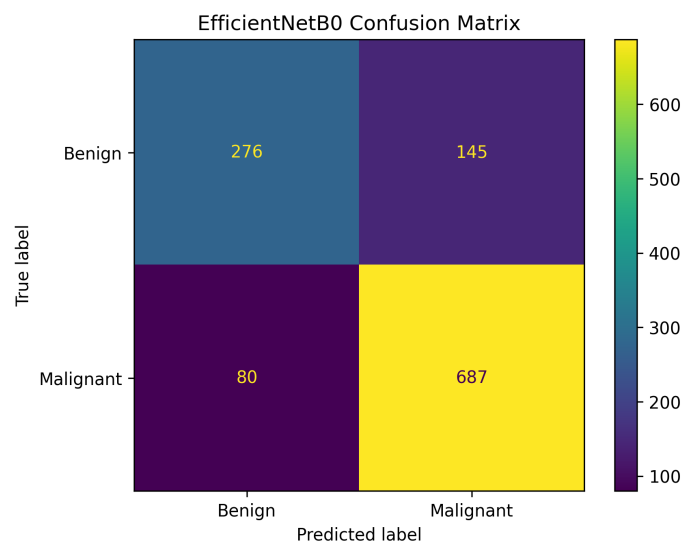


Figure 5.5: Confusion Matrix of EfficientNetB0

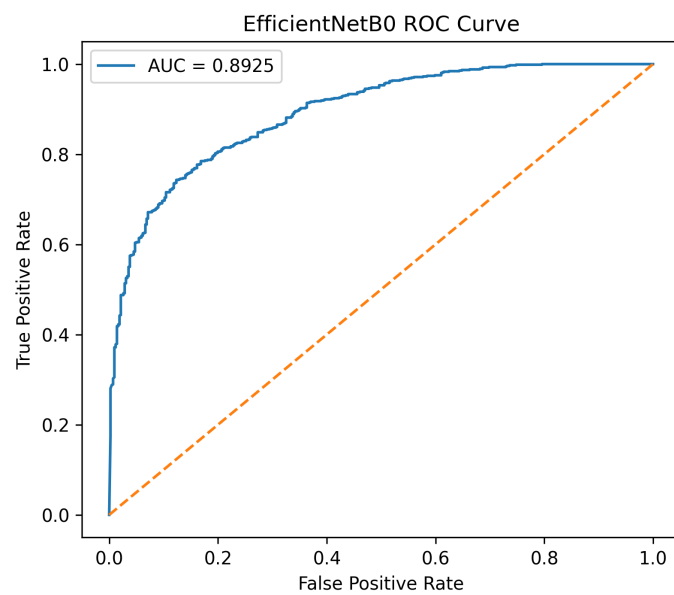


Figure 5.6: ROC Curve of EfficientNetB0

problem and also enhanced the learning of features.

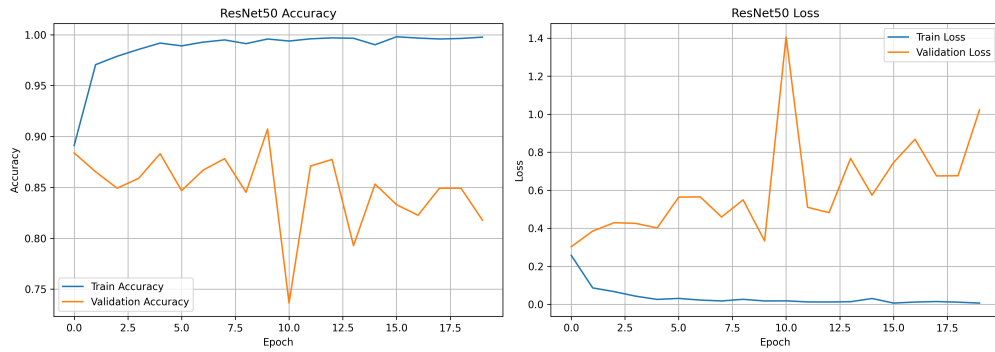


Figure 5.7: Training and Validation Accuracy/Loss Curves of ResNet50

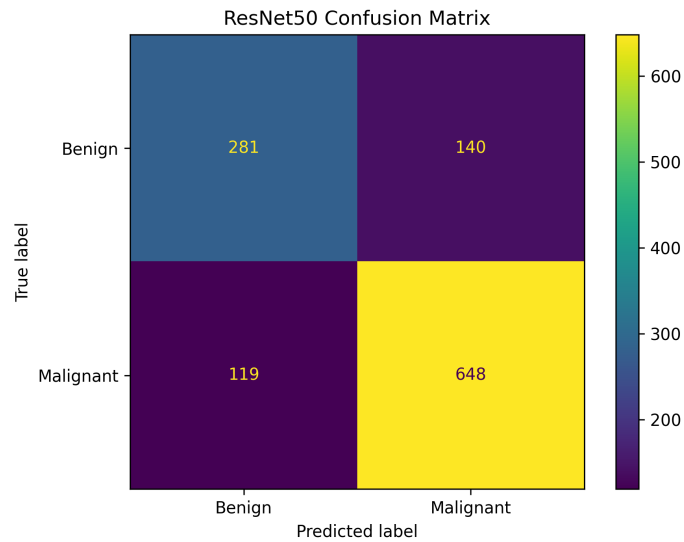


Figure 5.8: Confusion Matrix of ResNet50

5.4.4 Xception

Xception achieved the best overall performance with an accuracy of 81.48%, precision of 81.99%, recall of 91.40%, F1-score of 86.44%, and AUC of 88.79%. It demonstrated an excellent performance in extracting fine grain information in histopathology thanks to its “depthwise separable convolutions”.

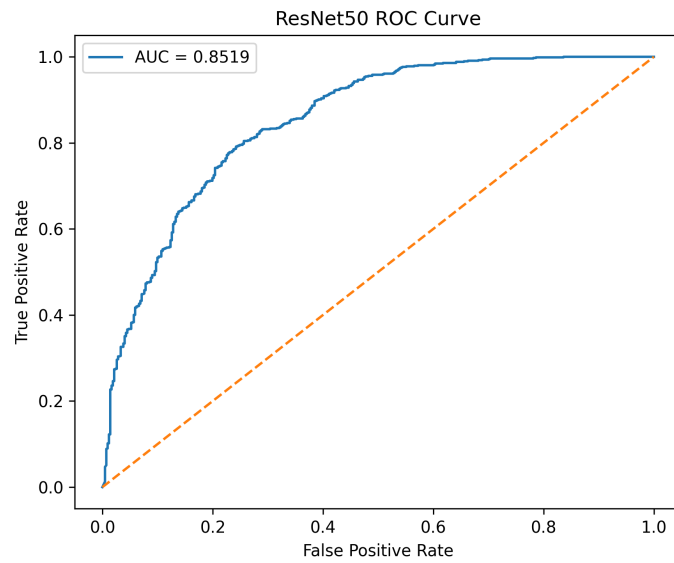


Figure 5.9: ROC Curve of ResNet50

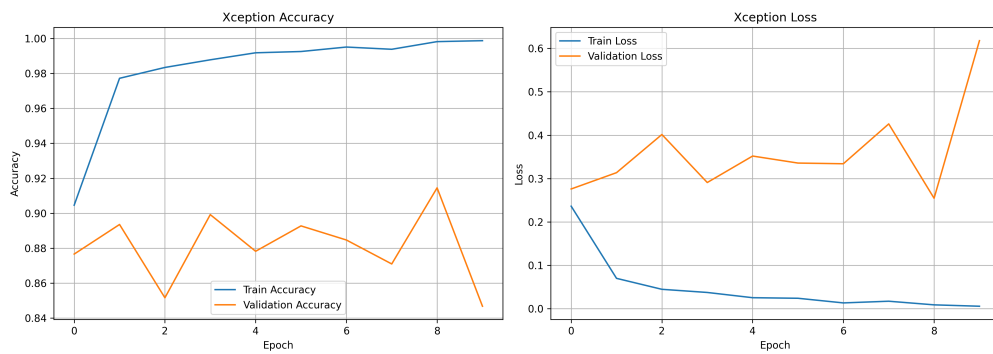


Figure 5.10: Training and Validation Accuracy/Loss Curves of Xception

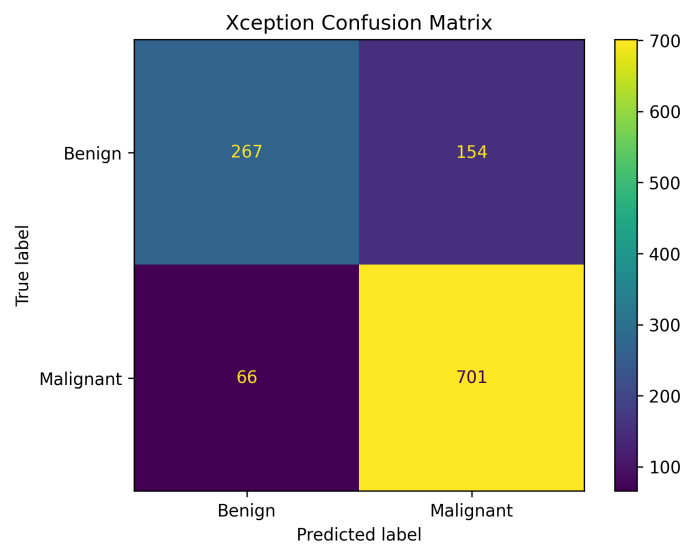


Figure 5.11: Confusion Matrix of Xception

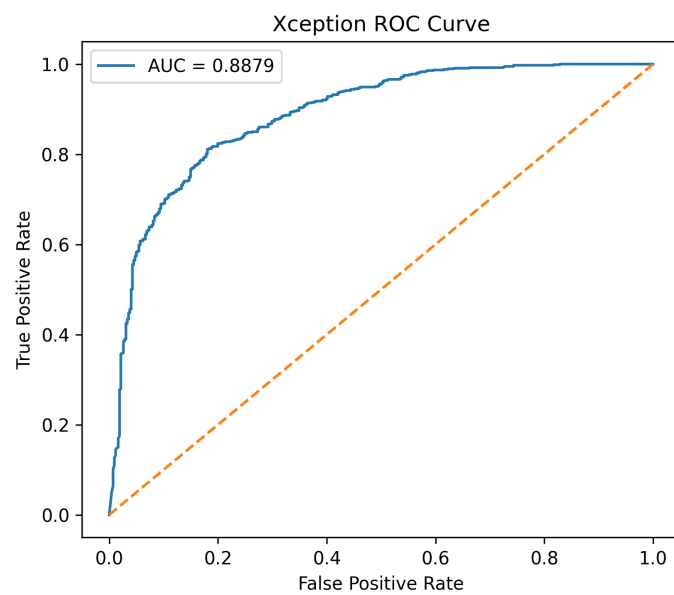


Figure 5.12: ROC Curve of Xception

5.4.5 MobileNetV3

MobileNetV3 achieved an accuracy of 80.05%, precision of 81.62%, recall of 89.18%, F1-score of 85.23%, and AUC of 86.25%. The lightweight architecture is applicable in medical image classification.

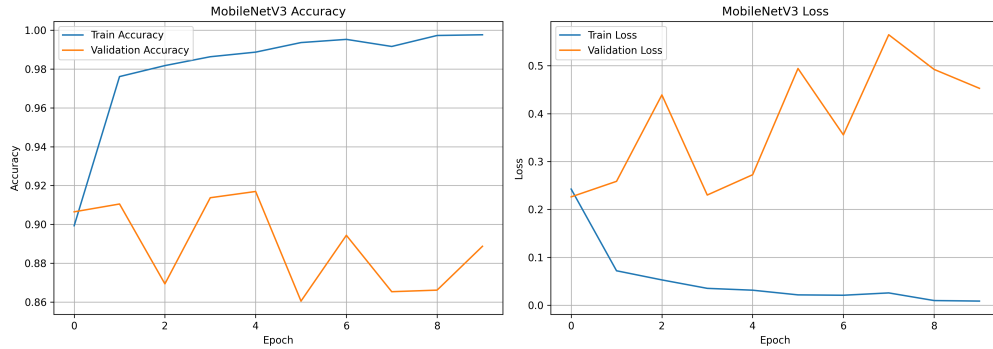


Figure 5.13: Training and Validation Accuracy/Loss Curves of MobileNetV3

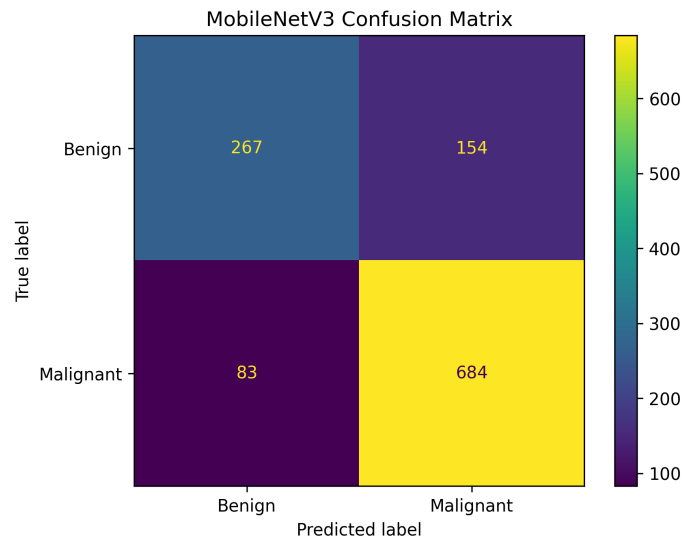


Figure 5.14: Confusion Matrix of MobileNetV3

5.4.6 Vision Transformer (ViT-B/16)

Vision Transformer (ViT-B/16) achieved an accuracy of 75.00%, precision of 75.60%, recall of 90.48%, F1-score of 82.37%, and AUC of 82.12%. While showing high recall rate, it was perform-

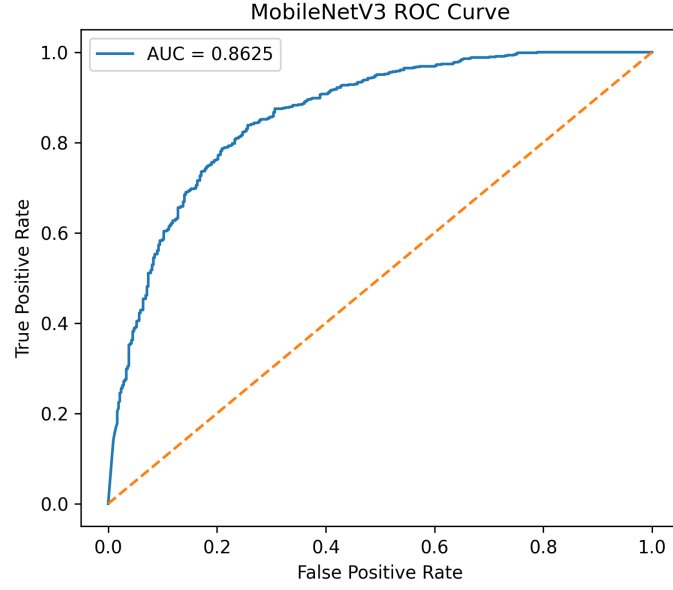


Figure 5.15: ROC Curve of MobileNetV3

ing lower in comparison with CNN based architectures may be because of the limited size of the data set.

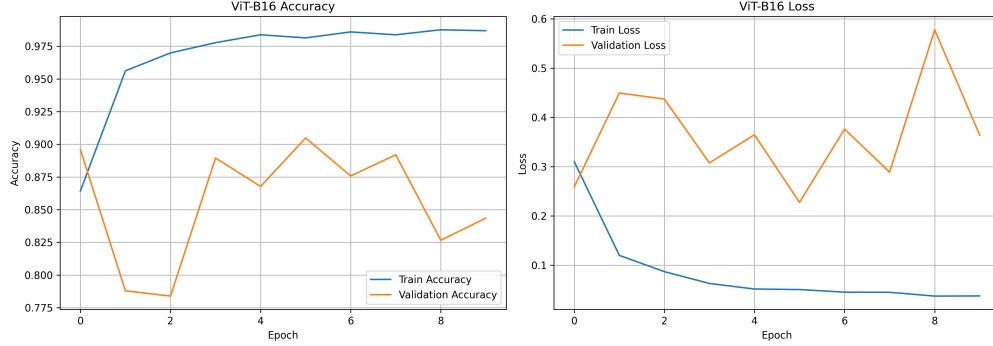


Figure 5.16: Training and Validation Accuracy/Loss Curves of ViT-B/16

5.5 Explainability Results

A model interpretability method is used to create grad-cam++ to improve the interpretability of the best performing model, MobileNetV3. The resulting heatmaps feature identified areas of the histopathological images, such areas with groups of nuclei or with abnormalities to the image, which also play an important role in the ultimate decision on the classification.

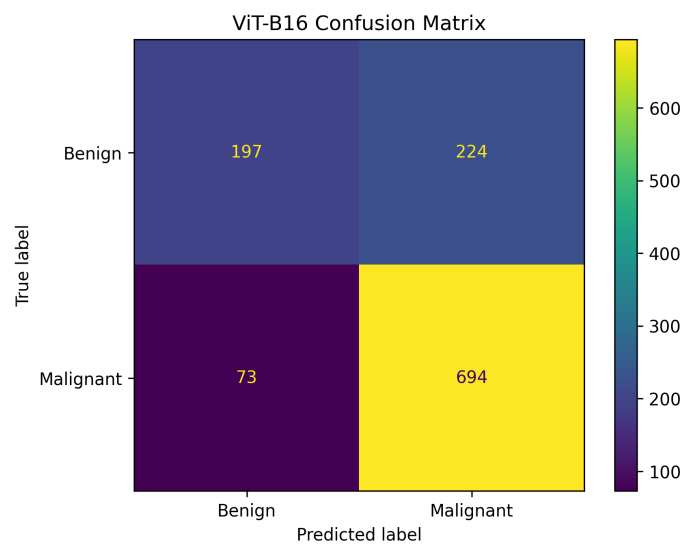


Figure 5.17: Confusion Matrix of ViT-B/16

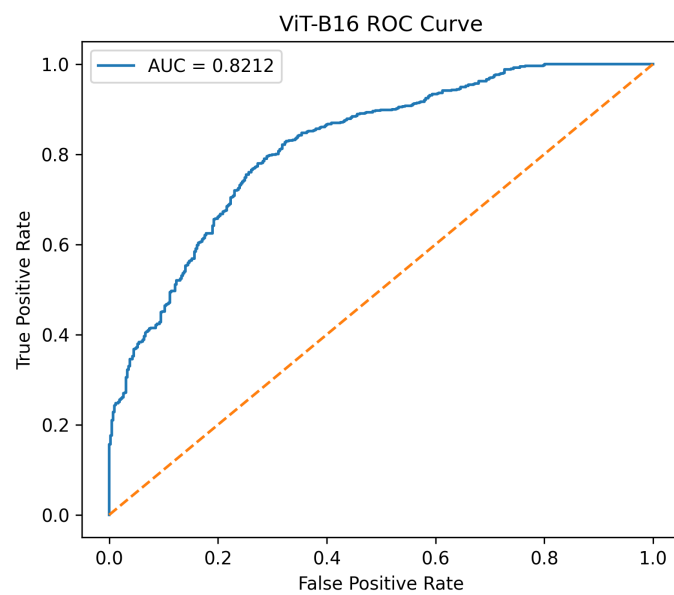


Figure 5.18: ROC Curve of ViT-B/16

These graphical explanations make the model more transparent, and make it more credible when predicting real-world results, as relevant to clinical applications.

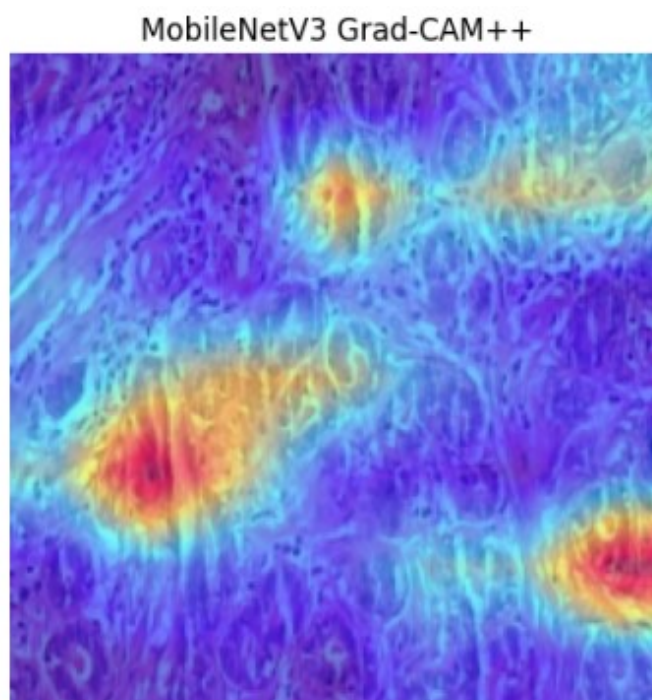


Figure 5.19: MobileNetV3 Grad CAM Plus

6 CONCLUSIONS

This paper provided a thorough comparative evaluation of deep learning and transformer-based models on the classification task for breast cancer using BreakHis histopathological image dataset. To create a precise and comprehensible automated system to differentiate benign and malignant tumour images of the breast. Thirty-two highly regarded architectures were tested within a consistent testbed: DenseNet121, EfficientNetB0, ResNet50, Xception, MobileNetV3 and Vision Transformer (ViT-B/16). Transfer learning and fine-tuning were used to train the models and several evaluation metrics, such as accuracy, precision, recall, f1-score and auc were used to evaluate the models. To evaluate the performance of the developed models on real-world algorithms, the experimental results showed that the model “Xception” was the best with highest F1-score and AUC and has performed well in capturing the discriminative features from the histopathological images. The dense connections in DenseNet121 allowed it to achieve high recall scores making it ideal for reducing false negative results in medical diagnosis. Currently, Vision Transformer (ViT-B/16) had competitive performance when evaluating on the recall metric, but when evaluating on the accuracy metric, this result was not as successful as it would be with the CNN-based architectures, which may be because the dataset was small. Overall, the performance of the convolutional NN networks was more stable and reliable for this task. In addition, the use of Grad-CAM++ enabled the model to be more explainable by making the area of interest in the histopathological images brighter when generating predictions. Besides, by highlighting the area of interest in the histopathological images, Grad-CAM++ gave a visual interpretation of the model’s outcomes, improving the model’s explainability. Increased clarity of the model and the greater trust in clinical decision making.

In conclusion, the proposed framework demonstrates that deep learning models, particularly Xception, can effectively assist in breast cancer diagnosis from histopathological images. The integration of explainable AI further strengthens the applicability of the system in real-world medical environments, supporting pathologists in early and accurate detection of breast cancer.

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